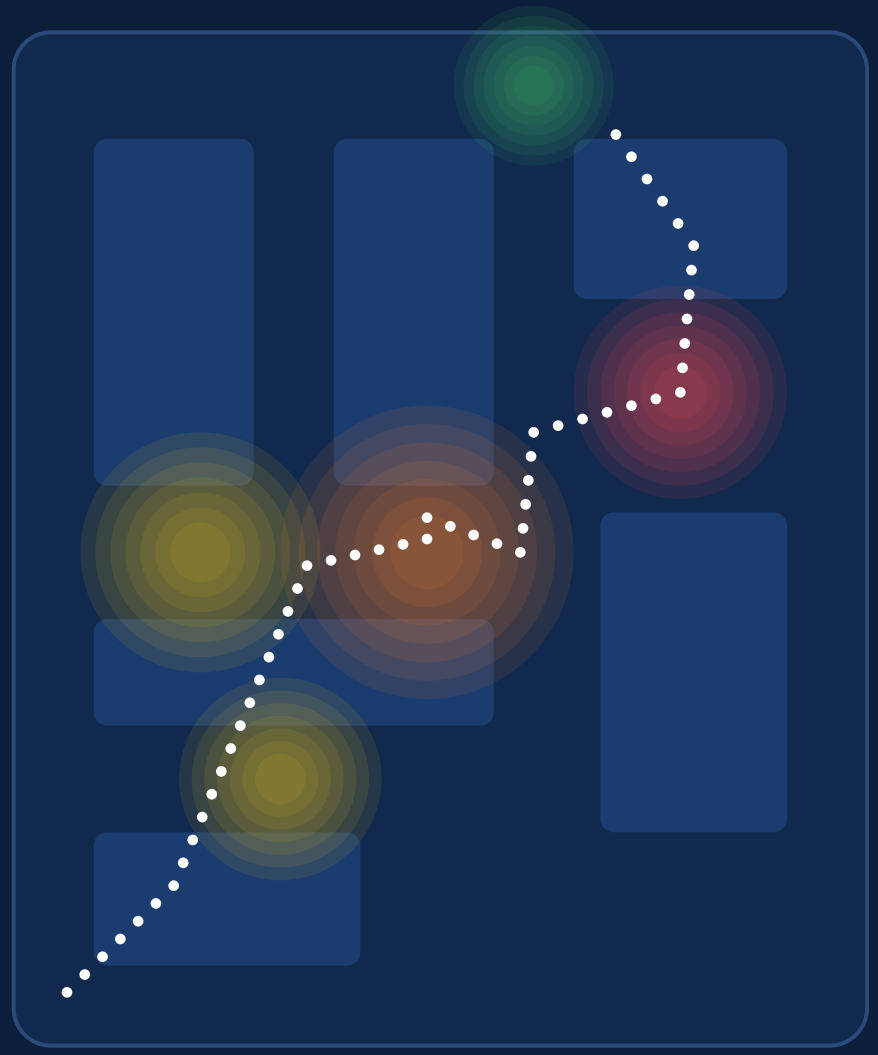
TEAM GRYZEN | IBM HACKATHON | PROBLEM STATEMENT 3

RetailDwell

Dwell-Time & Bottleneck Mapper

Online stores have analytics. Physical stores have guesswork.

We fix that with the cameras they already own, and tell managers whether a layout change worked and why.



Abhinav Biju | Rohitkumar S

Team Gryzen

THE PROBLEM

Stores know what sold, not why

POS

knows what sold

Door counter

knows how many came in

Everything in between

is guesswork

Sales data shows THAT a section underperforms. Each cause needs a different fix.

1 Nobody came

Visibility or placement problem

Move the display, fix the route

2 Came, did not stop

The display fails to attract

Redesign signage and sightlines

3 Stopped, did not buy

Price, stock or confusion

Fix labeling, stock, comparison

Who feels it: store managers, merchandisers, store planners, and suppliers who pay for display space. Mid-size retailers most: outgrown guesswork, cannot afford enterprise tools.

OUR DESIGN CONSTRAINT

We design for the hardest customer

The anonymous walker

Carries nothing.
Emits nothing.
Identifies nothing.

If we can serve this shopper from cameras the store already owns, every other sensor is a bonus.

Everything useful is derived from four observables:

1. **Where they are**
   Body position on the floor, over time
2. **How they move**
   Speed, direction, stopping, turning back
3. **Who they move with**
   Co-movement reveals shopping parties
4. **When they appear**
   Entrance and checkout timestamps

OUR SOLUTION

From raw CCTV to a decision

01

Existing CCTV

No new hardware. Camera agent turns footage into anonymous floor positions.

02

Zone analytics

Heat maps, dwell by zone, visit types, checkout queues, dead zones.

03

Experiment engine

Did the layout change work? Before vs. after, normalized for traffic.

04

Diagnostician

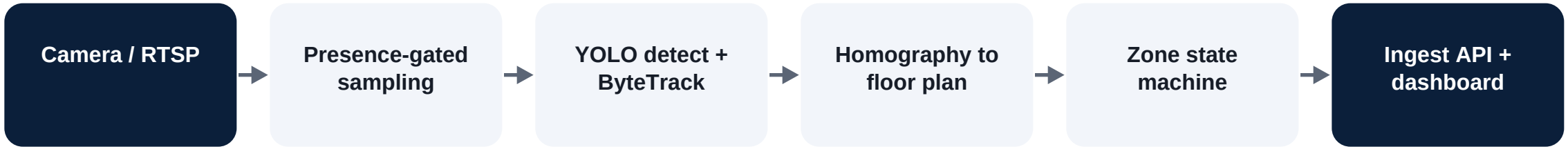
Why it happened and what to try next, in plain language.

Existing tools show a heat map. Few say whether a change worked.

We turn "we moved the display" into measured evidence a non-technical manager can act on.

HOW IT WORKS

How frames become floor coordinates



Sample smart

Frames stay in memory for one detection pass. Gate on person presence, not pixel change: a still shopper at a display IS the dwell.

Track reliably

YOLO with ByteTrack. Track IDs are random per run and never joined across runs. Tracks under 2 s are dropped so flicker never reaches the store.

Calibrate yourself

No CAD plan needed. Click 4 to 8 floor points you can measure with a tape; the agent reports the error in metres and draws a 1 m grid to check it.

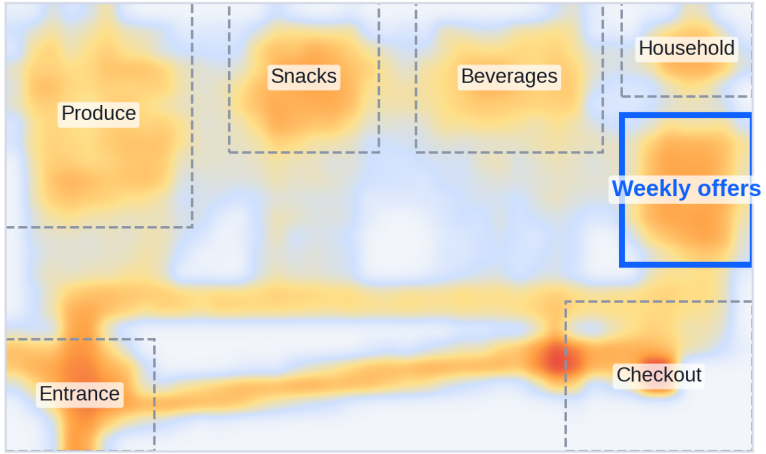
Honest status: agent, ingest API, analytics and dashboard are built. The agent is tested on synthetic video; **running it on real footage is the next milestone.**

THE SPINE: LAYOUT EXPERIMENT ENGINE

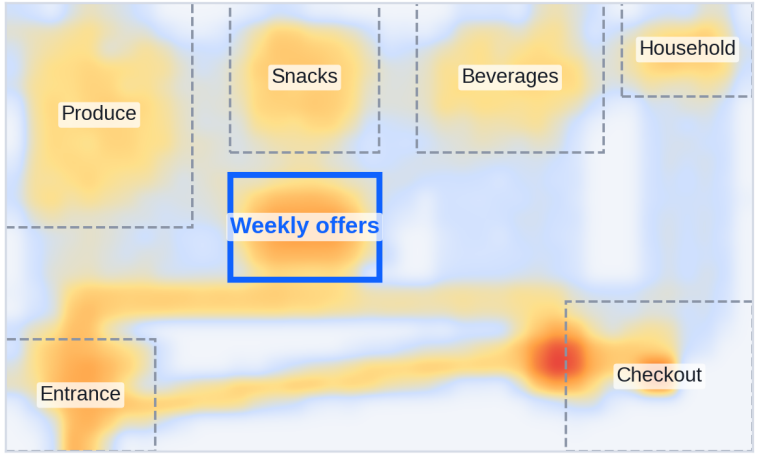
SIMULATED DEMO STORE

Did moving the display actually work?

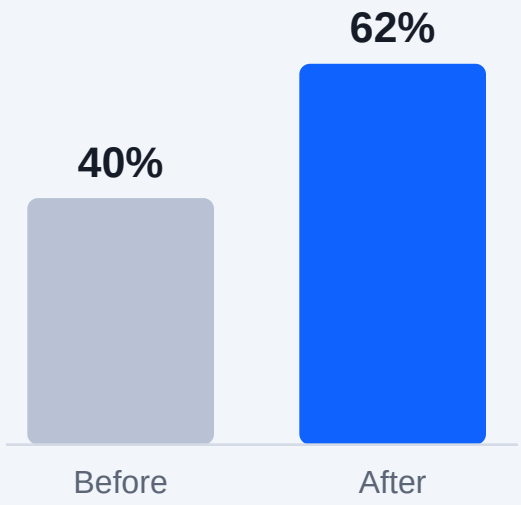
Before: offers on the far wall



After: offers beside the main aisle



Shoppers who stopped at Weekly offers



RAW STOPS AT THE OFFERS DISPLAY

+111%

130 to 274 stops. But 480 shoppers came on day B, against 350 on day A.

SHARE OF SHOPPERS WHO STOPPED

+54%

40% to 62% of shoppers. 95% range: +33% to +81%. Busy-day effect taken out.

Simulated store: we set the effect size, so this shows the engine recovers a known effect. It is not evidence about real shops.

ONE PIPELINE, FOUR STORES

SIMULATED DEMO STORES

It recovers known effects, and ignores non-effects

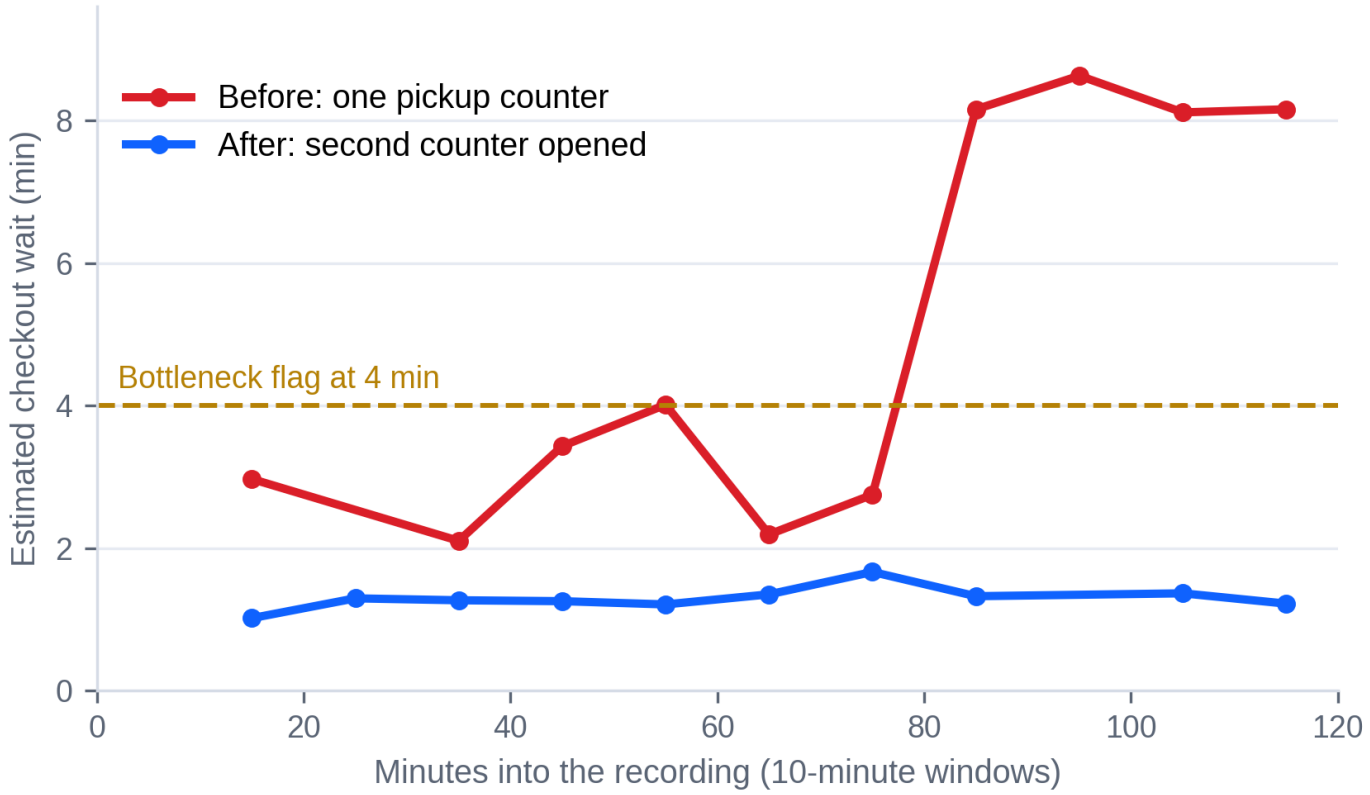
Store	What changed	Result
Lakeside Grocery	Offers display: far wall to main aisle	**Stopped share 40% to 62%. Lift +54% (95%: +33% to +81%)**
Volt Electronics	Accessories rack moved next to the tills	**Stopped share 33% to 53%. Lift +63% (95%: +34% to +98%)**
Thread & Co.	New-arrivals table moved to the entrance	**Stopped share 34% to 63%. Lift +86% (95%: +56% to +123%)**
Corner Pharmacy	Second pickup counter opened, same floor plan	**Median wait 3.7 to 1.3 min. Worst window 8.6 to 1.7 min**
Control	Pharmacy "Everyday health": untouched	**Reads as no effect: -6% (95%: -23% to +12%)**

Why stopped share is the headline: a display moved onto a main path is walked past more, which flatters a plain reach count (Volt: reach +176%). Stopping is what attention looks like.

LIVE OPERATIONS

SIMULATED DEMO STORE

Bottlenecks: the queue, caught early



Ignores the first 10 minutes (warm-up) and windows serving under 0.8 people per minute.

MEDIAN ESTIMATED WAIT

3.7 min to 1.3 min

WORST 10-MINUTE WINDOW

8.6 min to 1.7 min

A queue is not a crowd

Wait = people in the queue area divided by people served per minute (Little's law). Flag at 4 minutes or more. It assumes steady state, so treat it as indicative during surges.

SUPPORT: THE DIAGNOSTICIAN

Every flag explains itself

EXAMPLE FLAG, OUTPUT FROM THE DEMO (LAKESIDE GROCERY)

Household: visits 69% below the store median, dwell normal. People are not seeing it. Move signage to sightline height or closer to the main path.

What the cameras see	Flag	Suggested action
Visits under 0.7x store median, dwell not low	**Visibility problem**	Raise signage, move closer to the main path
Visits and dwell both under 0.7x	**Dead zone**	Relocate or redesign the display
Visits over 1.3x, dwell under 0.7x	**Pass-through**	Put impulse items and promos here
Visits and dwell both over 1.3x	**Hot zone**	Protect its footprint
Visits not low, dwell over 1.3x	**Destination**	Add seating, demos or staff
Estimated checkout wait of 4 min or more	**Bottleneck**	Open another counter

The store-wide median stands in for a baseline. Each zone is also checked against its own time-of-day norm (30-minute windows, flagged only when unlikely to be chance).

TRUST

Privacy by design, not by policy

~300 GB

of video per week (typical, from our ideation doc)

becomes

~10 MB

of anonymous coordinates per week

`(track_id, t, x, y)`

Our estimate for 1,000 visits a day, from simulated visits: about 200 points each, about 7 bytes per point compressed.

No faces, no stored video

Frames are processed in memory and discarded. Nothing but four fields leaves the machine.

Four fields, enforced in code

The schema has no other column. CSV upload drops extras; the ingest API rejects them.

Short-lived IDs

Random per run, never joined across runs. No repeat-visitor identity, no phone IDs.

Locked down access

Row-level security in Postgres. API keys shown once, stored as a SHA-256 hash.

Aggregates only

No alerts about individuals. This is analytics, not surveillance.

ACCURACY

What we measured, and what we have not

MEASURED

- Position error on synthetic video: median 0.09 m, 90th percentile 0.22 m (no-model detector).
- Same test: 73% recall, 99% precision, 45 identity switches (29 tracks for 14 people).
- Engine recovers known effects and returns no effect on a control zone (4 simulated stores).
- 19 of 19 agent tests pass, video to ingest API to analysis.

PROPOSED TARGETS

- At least 90% of annotated stops land in the correct zone.
- Median dwell error within 20% of the annotated value.
- Entrance and checkout counts within 10%.
- ID switches reported per minute at the chosen frame rate.

NOT YET MEASURED

- Dwell error, count error and ID switches on real footage.
- The YOLO path on real clips (tested only against a fake model so far).
- Mission and loiterer labels on scripted clips.
- Journey completeness, entrance to checkout.

Honest notes

Accuracy will be stated at zone level, not centimetre level. Path to purchase is correlation, not causation. All demo numbers are simulated, and the analysis has not yet seen real footage.

WHERE WE FIT

Affordable, private and actionable

	Typical enterprise sensor platforms	RetailDwell
Hardware	Purpose-built overhead sensors	**Cameras the store already owns**
Store map	CAD or PDF plan from the retailer	**Self-serve calibration, no blueprint**
Output	Heat maps and counts	**Experiments plus plain-language diagnosis**
Built for	Chains with rollout budgets	**Mid-size retailers**

Who pays

Mid-size retailers that have outgrown guesswork but cannot afford enterprise tools. A subscription.

Who else pays

Suppliers who buy display space and want proof it is seen. Our before/after lift is that proof.

Comparison reflects typical vendor approaches from public material, not an exhaustive or verified feature check. Business model is an idea, not yet validated.

WHERE WE ARE

Built today, and what comes next

BUILT AND WORKING

- Heat maps on a floor plan; zone stats: reach, presence dwell, engaged dwell, stop rate
- Visit types (directed, exploratory, loiterer, pass-through) with confidence
- Plain-language flags, plus time-of-day baselines per zone
- Layout experiment engine: traffic-normalized, 95% bootstrap range
- Checkout wait with Little's law, and a bottleneck flag
- Accounts, read-only demo, floor plan upload, zone editor, CSV upload
- Ingest API with per-store keys, and the camera agent

NEXT, NOT BUILT YET

- Validate the agent on real footage
- Multi-camera fusion
- POS matching and revenue per visitor
- Layout what-if simulation
- Google sign-in

Stack: Python, Flask, Supabase (Auth, Postgres with row-level security, Storage), pandas, NumPy, Plotly.js, Ultralytics YOLO, OpenCV.

RETAILDWELL | TEAM GRYZEN

Stores already own the cameras.
We turn them into an experiment lab.

Measures

Did the layout change work? A traffic-normalized lift with a 95% range.

Explains

Plain-language flags for dead zones, visibility problems and queues, each with a fix.

Respects privacy

Four anonymous fields per detection. No faces, no stored video.

CODE AND DEMO

github.com/AbhinavBiju-101/RetailDwell

```
pip install -r requirements.txt   &&   python app.py
```

Runs locally with a demo account and four simulated stores. Setup steps are in the README.

Scan to open

the repository with the code and the demo.

Team Gryzen | Abhinav Biju | Rohitkumar S